

The TECHNICAL DESIGN:

1. First of all, I am implementing a CLI based interface for this project.
2. So, how it works is? A user can click on `main.py` file in the project folder and it prompts you to enter the relevant details (like an individual's tuple from the Recruiter CSV export or linkedin URL or github URL, and many more) and the config file is considered, following that, the parsing (we will discuss about it in detail) is done and the output is stored in `candidates` directory that to in .json file.
3. The REASON I DECIDED to store the processed candidate details in .json file offline is to make the processed data persistent.
4. The details are normalized as asked in the document like phone number as E.164 format and many more.

The overall pipeline looks like:

Ingest -> per source adapters -> normalize -> entity resolution -> merge -> confidence scoring
-> project (apply config) -> validate -> emit

5. The REASON I DECIDED to make the parsing modular for each source is to save the relevant sources (contains correct information about a candidate) from irrelevant sources (what if user uploaded some random file instead of intended file).
6. RULE TO MERGE: Two records are belongs to the same candidate if,
 - a. normalized email matches, OR
 - b. at least 2 of {phone, github handle, linkedin handle} match (all normalized first ie lowercased email, E.164 phone, stripped-down handle from URL or raw username).
 - c. Otherwise, treat them as separate candidates, full stop, no flagging, no in-between state.
7. The REASON I DECIDED not to treat the details like phone, github handle and linkedin handle as important as email is same phone number can be assigned to someone else if phone number becomes dormant. Similarly, if the user changed this username, platforms like github and linkedin will assign the same username to someone else after certain period of time.
8. One of the possible edge cases could be, what if there's no country code provided in the phone numbers? For those, the confidence of that field is affected.
9. How I handle the runtime config file?. So, when we parse the sources, we are storing everything we get from the sources. But, the config file only affects what get projected out of it. That's it.
10. Also, all the details that we extract from the sources are normalized and stripped from the both ends, so that, the comparisons made are fair and correct.
11. The confidence is calculated using the formula:
$$\text{overall_confidence} = \text{avg}(\text{populated field_confidences}) \times \text{identity_match_factor}$$

Where field confidences are calculated separately, ie suppose there's a field called phone number, if the phone number doesn't has country code, then that field will have less confidence than the one with country code.

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